

How memory is allocated to ENUM

What is an optional class, when do we use it.
What is stream API? Why do we require the Stream API
Difference between Collection API and Stream API
What is the difference between Map and flatMap Stream Operation
Explain about lazy loading in Hibernate
How transaction management is done in your application, Explain the various ways to perform transaction management.
How exception are handled in microservices.
Micro service is deployed on different machine and on one machine the service fails randomly, how do you identify that machine
What problem does microservice solved. What problem you face in monolithic application design.
What difficulties you have faced while developing or deploying the microservices.
What do you mean when we say rest apis are stateless and what is the advantages or disadvantages of it.
What are the different methods in REST explain when to use what

GitHub Copilot

Explain the very technical code to non-technical users like managers and senior VPs
Please keep responses short
Please keep responses formal
Management and non-technical users
Plain text

https://chromewebstore.google.com/detail/joycode/mihgbgcojjcgckpoclhjoneafcmfpng?utm_source=ext_app_menu
https://bmwgroup.sharepoint.com/:p:/t/x_labamericas/EfleyB6t1PJHudbVa4MwJOkBjLsdLG29yZQPRd_w8vTadQ?e=b9bo0K
<https://atc.bmwgroup.net/confluence/display/AI4DEVOPS/AI4DevOps>
<https://joycode.bmwgroup.net:8440/topic/category/Best%20Practice>
<https://atc.bmwgroup.net/confluence/pages/viewpage.action?pageId=5611682895>
<https://joycode.bmwgroup.net/#/home>
<https://joycode.bmwgroup.net/#/installation-guide?key=vscode>
<https://marketplace.visualstudio.com/items?itemName=BMWLingYueCIH.JoyCode>
How to Start Hands-On_JoyCode
Dev Use Cases Demo_JoyCode
JoyCode Introduction.mp4
Event Introduction_Frank_JoyCode *
OPS USE CASE DEMOS

Team Name: [Team or Role Name]
Purpose: [Brief description of the team or role's purpose]
Key Products/Services: [List of key products/services]
Key Stakeholders: [List of key stakeholders]
Common Acronyms: [List of acronyms with definitions]
Examples: [Brief example(s) of projects or interactions]

The below information describes my team and what we do.

Team Name: DTO Americas (Data Transformation Office Americas Region)
Purpose: Data Transformation topics including Data Management and Data Governance enablement and Training. The goal is to enable Data as a Product for use and reusability.

- Key Products/Services: Data Governance Framework awareness, Consulting, Training, Onboarding
- Key Stakeholders: Business Use Case Owners in various Domains, Data Governors, Data Stewards, Data Engineers.
- Common Acronyms: DMGF
- Examples: [Brief example(s) of projects or interactions]

=====

The below information describes my team and what we do.

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Using the milestones, can you summarize the categories of interactions, projects, engagements and tasks for the team this year?

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Do I have any new tasks or requests from Confluence or jira today?

Do I have an emails requiring urgent attention? Can you help me prioritize?

Do I have new meetings for today?

=====

<https://outlook.office.com/mail/Anupkumar.Singh@partner.bmw.in@bmwgroup.mail.onmicrosoft.com>

[https://atc.bmwgroup.net/confluence/display/DTOAMX/2024+-+Milestones DTO Americas Region](https://atc.bmwgroup.net/confluence/display/DTOAMX/2024+-+Milestones+DTO+Americas+Region)

MFG130 COMPLIANCE

dwni6u5x8uyyf.cloudfront.net/awm-governance7

gaia.bmwgroup.net/genai/dev-space/edit-app/e5dca2f9-770e-46f6-97bf-24b6d4e48215
(https://atc.bmwgroup.net/confluence/display/GENEXT/Guidance+on+requesting+GAIA+RightN
ow+roles)
gaia.bmwgroup.net/genai
https://mail.bmwgroup.net/owa
Welcometobmwindia2020#2024
qxz1mm1@asiapacific.bmw.corp/pswd
Gwendolyn.Stamp

turning.com/interview-questions/graphql
hasura v2.0.6
graphql
http://10.5.198.157:31176/v1/graphql
Liquibase

cd C:\Users\qxz1mm1\jupyter
jupyter notebook

https://youtu.be/p3Alecylvok4 Hosting Spring Boot App on render.com
https://youtu.be/q8OleYuqntY https://www.techworld-with-nana.com/
nginx.org/en/docs
SSL/TLS
https://youtu.be/1jCM4jKV68U

Amazon
Lex, Polly, Rekognition, ML

server:
port: 8443
ssl:
Accepting HTTPS connections only
enabled: true
The path to the keystore containing the certificate
key-store: classpath:keystore.p12
The password used to generate the certificate
key-store-password: tomcat
The format used for the keystore
key-store-type: PKCS12
The alias mapped to the certificate

key-alias: tomcat

server:

ssl:

key-alias: spring-https

key-store: classpath:javatechie.jks

key-store-type: JKS

key-store-password: password

port: 7777

spring-https/src/main/resources/application.yml

keytool -genkey -alias spring-https -storetype JKS -keyalg RSA -keysize 2048 -validity 365
-keystore javatechie.jks

last name: You tube

org unit: Technology

org: Javatechie

City: Bangalore

State: Karnataka

Country: IN

Warning:

The JKS keystore uses a proprietary format. It is recommended to migrate to PKCS12 which is an industry standard format using

"keytool -importkeystore -srckeystore javatechie.jks -destkeystore javatechie.jks -deststoretype pkcs12".

<https://platform.openai.com/tokenizer>

<https://docs.spring.io/spring-framework/reference/web/webmvc-view.html>

<https://www.luv2code.com/bonus-lecture-spring-boot-rest-security-jpa-hibernate-bcrypt-pdf>

<https://www.luv2code.com/bonus-lecture-spring-boot-rest-security-jpa-hibernate-bcrypt-code>

<https://www.bcryptcalculator.com/>

<https://luv2code.com/generate-bcrypt-password>

<https://luv2code.com/why-bcrypt>

(<https://danboterhoven.medium.com/why-you-should-use-bcrypt-to-hash-passwords-af330100b861>)

<https://luv2code.com/bcrypt-wiki-page> (<https://en.wikipedia.org/wiki/Bcrypt>)

<https://luv2code.com/password-hasing-best-practices>

(<https://crackstation.net/hashing-security.htm>)

C:\Users\qxz1mm1\Downloads\psql\bin>pg_ctl -D C:\Users\qxz1mm1\psql\mydata -l logfile

start

<https://youtu.be/cYFYfYXObgA>

Grafana : Setup Grafana for Spring Boot app | Actuator, Prometheus & Grafana | Monitoring & Alerting

<https://youtu.be/gJZhdEJvZmc>

Filters

HttpBasicAuthentication(Authentication Filter) (FilterChain, SecurityContext)

AuthenticationMgr

AuthenticationProvider(Multiple Auth Prvd) uses (UserDetailsService, PasswordEncoder)

Intro

Dependencies

Lifecycle, goal, plugins

Archetypes

.m2e eclipse plugins

Intro /benefits

Major component

Component Interact

Tool install

Hello world demo

Artifact version

Change logs

Documentation

Javadocs

Reports

<directory>

C:\Users\qxz1mm1\eclipse-workspace\more-moven-example\production

</directory>

<profiles>

<profile>

<id>production</id>

<activation>

<property>

<name>env.PACKAGE_ENV</name>

<value>PROD</value>

</property>

</activation>

</profile>

</profiles>

mvn dependency:copy-dependencies

<https://maven.apache.org/guides/introduction/introduction-to-the-standard-directory-layout.html>

super pom

maven\apache-maven-3.2.5\lib\maven-model-builder-3.2.5.jar\org\apache\maven\model\

mvn help:effective-pom

mvn archetype:generate -DgroupId=com.mycompany.app -DartifactId=my-app

-DarchetypeArtifactId=maven-archetype-quickstart -DarchetypeVersion=1.5

-DinteractiveMode=false

<scope>test</scope>

<scope>compile</scope>

mvn -X compile

mvn -X test

====phase in life-cycle====

mvn help:describe -Dcmd=clean

mvn help:describe -Dcmd=deploy

mvn clean install

compile

Test-compile

Test

Package

Install

Deploy

mvn compiler:compile

mvn help:describe -Dplugin=compiler

mvn compiler:help -Ddetail=true -Dgoal=compile

mvn compiler:compile -Dmaven.compiler.verbose=true

mvn archetype:create -DgroupId=com.infiniteskills.maven -DartifactId=first-custom-plugin

-DarchetypeGroupId=org.apache.maven.archetypes

-DarchetypeArtifactId=maven-archetype-mojo

mvn install

mvn com.infiniteskills.maven:first-custom-plugin:touch

<https://youtu.be/uAQs-YXnY-U>

dependencies(3rd party lib/maven local repositories, db,selenium,test),

plugins(proj configuration/compiler/run)

maven remote repositories(mvn repositories)

report/documentation

packaging

validate/Compile/Test/Packing/Integration/Verify/Install/Deploy
<https://mvnrepository.com/TestNG/mysql>

<https://automationstepbystep.com/jmeter-beginner/>

mapReduce

fork and join

return Objects.hash(name,age);

if(this == o) return true;

if(o == null || getClass() != o.getClass()) return false;

Person person = (Person) o;

return name.equals(person.name) && age==person.age;

1. Given a sentence, find and print the frequency of each word.
2. Given a list of integers, find out all the numbers starting with 1.
3. Given a list of names, group them by their first letter, and then count the number of names in each group.
4. Find and print duplicate numbers in an array if it contains multiple duplicates?
5. How are duplicates removed from a given array in Java?
6. Given a list of words, filter and print the palindromes
7. How do you merge two sorted arrays into a single sorted array?
8. Given two lists of strings, concatenate them and remove duplicates.
9. Student Grade Classification - 70 and above pass
10. Given a list of strings, sort them according to increasing order of their length.
12. Partition a list of numbers into two groups: even and odd, using a custom predicate.
13. Find the squares of the first three even numbers in a list.
14. Flatten a list of lists

<https://github.com/VarshaDas/Java-Code-Snippets/blob/main/LambdasAndStreams/src/main/java/StreamsCodePart2>

<https://youtu.be/1gXFrh9PBic?list=PLUDwpEzHYYLvTPVqVlt7tIBohABLo4gyg>

<https://youtu.be/HwwXiCf-nOM>

You

3:19 PM

Kadane's algorithm

Hover over a message to pin it

keep

key clock/ active directive

Akshay Rastogi

3:28 PM

<https://medium.com/@itsmynameangad/java-build-your-own-custom-executorservice-threadpool-213e46159326>

<https://medium.com/codex/executorservice-internal-working-in-java-7b286882f54e>

santosh kumar

3:45 PM

```
private static void maxProfit(int[] prices) {  
    int maxProfit = 0;  
    int buy = prices[0];  
    for (int j = 1; j < prices.length; j++) {  
        if (prices[j] > buy) {  
            maxProfit = Math.max(maxProfit, prices[j] - buy);  
        }  
        else {  
            buy = prices[j];  
        }  
    }  
    System.out.println(maxProfit);  
}
```

Akshay Rastogi

4:03 PM

Pow(x, n)

Medium

Valid Perfect Square

a=[5,2,6,1,4]

a=[3,5,1,7,4,9,3]

a={3,1,4,8,7,2,5}

Thread Pool executor

<https://medium.com/@itsmynameangad/java-build-your-own-custom-executorservice-threadpool-213e46159326>

<https://medium.com/codex/executorservice-internal-working-in-java-7b286882f54e>
function.identity() in java

Kadane's algorithm

<https://www.youtube.com/watch?v=5T0SiJocPCI>

<https://leetcode.com/problems/maximum-profit-in-job-scheduling/description/>

hackerearth

leetcode

codeforces

Linear and Binary Search

https://youtu.be/5h_wcuRI7kE

<https://youtu.be/xxVwdMwHPqg> Builder Design Pattern for Beginners with easy examples | Low Level System Design

<https://youtu.be/t2HmF3Y4WgY> Factory Design Pattern | Abstract Factory Pattern for Beginners with easy examples

<https://www.javatpoint.com/abstract-factory-pattern>

<https://youtu.be/7g9S371qzwM> DIFFERENCE BETWEEN FACTORY AND ABSTRACT FACTORY

STATE DESIGN PATTERN https://youtu.be/wOXs5Z_z0Ew

Behavioural pattern

<https://www.youtube.com/watch?v=DBDnUkTobaE&t=252s>

State Pattern: allows an object to alter its behavior when its internal state changes (Wending Machine)

allows an object to alter its behavior when its internal state changes.

Product Has-a State (Interface) Is-a (inheritance) State 1 State 2 State 3

State productState

Observer Pattern: in this an object(Observable) maintains a list of its dependents (observers) and notifies them of any changes in its state

Strategy Pattern: help to define multiple algorithms for the task and we can select any algorithms depending on the situation PayStrategy (CreditCard/ UPI Payment/ Cash Payment

Chain of Responsibility: allows multiple objects to handle a request without the sender needing to know which object will ultimately process it.

Template Method Pattern: when you want all classes to follow specific steps to process the tasks but provide flexibility that each class can have their own logic in that specific step.

Interpreter Pattern: Defines a grammar(CONTEXT) for interpreting and evaluating an expression.

Command Pattern: Turns requests (commands) into objects, allowing you to either parameterized or queue them

This will help to decouple the request Sender and receiver.

Iterator patterns provide a way to access elements of a Collection sequentially without exposing underline representation of the collection.

Visitor Pattern: Allows adding new operations to existing classes without modifying them and encourage OPEN/CLOSE principle.

Mediator Pattern: It encourage loose coupling by keeping objects from referring to each explicitly and allows them to communicate through a mediator object.

Memento Pattern: Provides an ability to revert an object to previous state i.e. UNDO capability and it doesn't

HOW TO REMEMBER

<https://www.codeproject.com/Tips/57578/The-best-way-to-remember-design-patterns>

DESIGN PATTERN

https://youtu.be/ASI0TfcY_7U SINGLETON DAILY CODE

<https://youtu.be/zABYFmRs6No> BUILDER PATTERN

QUEUE <https://youtu.be/PvDoT79oHTs>

STACK <https://youtu.be/fptlqsesjxY>

Cache
App Data
DB

Read Through
Write Through
Write Ahead DB update Cache
Write Behind

Master Worker
Segregation pattern

Push the msg in Queue

<https://youtu.be/4aYvLz4E1Ts> java Thread
<https://engineeringdigest.medium.com/multithreading-in-java-39f34724bbf6>

<https://www.sql-practice.com/>
<https://youtu.be/vz6vSZRuS2M> Java Memory Management and Garbage Collection in Depth
Topics which i have covered:

=====

Two type of memory - STACK , HEAP Manage by JVM
STACK - temp variable (defined inside a block), primitive type,
reference of object.
Each thread have separate STACK memory.
StackoverflowError - if stack memory full.

Stack

memo block
[

strLit
personObj
primVar =10
]

Heap

[
string pool [24]

```
MemoObj()  
Person()  
]
```

minor GC, major GC, age Threshold

Yng Gen, S0, S1 (Survivor space) / Old Gen, permgen(part of heap and not expandable) (Meta space)

Non Heap (Meta space) - Class variable (static), Class metadata (Info abt class), Constant, Meta data is expandable

Mark & Sweep

Mark & Sweep with Compact (arrange)

Serial GC

Parallel GC

Concurrent Mark & Sweep

G1 was first introduced in Java 7, but became the default in Java 9

Yes, Epsilon GC is available in JDK 11:

What it is

Epsilon GC is an experimental, no-op garbage collector that allocates memory but doesn't perform any memory reclamation.

When to use it

It's designed for use cases where garbage collection is not required or can be eliminated, such as for performance testing, short-lived applications, or applications that manage memory explicitly.

How to use it

To use Epsilon GC, you need to specify the following two runtime switches on the command line:

-XX:+UnlockExperimentalVMOptions

-XX:+UseEpsilonGC

Yes, Shenandoah GC is available in Java versions 8, 11, and 12 and later:

OpenJDK 8: Shenandoah GC is available as an experimental feature

OpenJDK 11: Shenandoah GC is available as an experimental feature

OpenJDK 12 and later: Shenandoah GC is available as a production feature

Type of Memory (Stack and Heap)

What kind of data is Stored in Stack and Heap with Example

Types of References

Strong Reference

Weak Reference

Soft Reference

Heap Memory Structure

Young Generation

Eden

Survivor (S0 and S1)

Old Generation

Metaspace

How Garbage Collector work & clean up the Heap memory with Example

Types of Garbage Collector

Single GC

Parallel GC

CMS (Concurrent Mark and Sweep)

G1 GC

<https://youtu.be/QovOdd80A4s> TIME COMPLEXITY

<https://youtu.be/aOen1-pQLZg>

<https://youtu.be/6O5iJ7PKUhs>

Microservices Architecture Patterns | SAGA Choreography Explained & Project Creation | JavaTechie

Spring WebFlux

Apache Kafka

Spring Cloud Stream

Java 8

Multi Module project:/ Reuse component classed

Orchestration Design Pattern | Distributed Transaction Management| MicroService Design Patterns

<https://www.youtube.com/watch?v=J3sh3leLFh0>

<https://youtu.be/YGnpHZcuMOo>

How to immutable class

sync method call sleep will give other thread waiting thread will execute or not.

creational structural behavioural (3 example of each)

Best/Avg/worst time complexity/space complexity of Array/ Linked List. Write it.

Difference between Error/ Exception / Checked/ Unchecked / Custom Exception is checked/uncheck.

why Stram API when we have Collection

how singleton work in case of serialization/deserialization

It seems you might be referring to Java's Integer::sum method. This method is used to sum two individual integers, not a list of numbers.

To sum a list of numbers in Java, you can use streams and the collect() terminal operation:

```

import java.util.List;
import java.util.Arrays;
import java.util.stream.Collectors;

public class SumList {
    public static void main(String[] args) {
        List<Integer> numbers = Arrays.asList(1, 2, 3, 4, 5);

        int sum = numbers.stream()
            .collect(Collectors.summingInt(Integer::intValue));

        System.out.println("Sum of the list: " + sum);
    }
}

package com.bmw.beamapis.exam;

public class MainTest {

    static {
        System.out.println("static block");
    }

    MainTest(){
        System.out.println("Constructor");
    }

    public static void main(String[] args) {
        MainTest m = new MainTest();
        System.out.println("main method");
    }

    {
        System.out.println("initialize");
    }
}

```

A.
static block

main method

B.

static block

initialize

Constructor

main method

Choose from the below options

```
public class LoggerExample {  
    private StringBuilder contents = new StringBuilder();  
    public void log(String message){  
        contents.append(System.currentTimeMillis());  
        contents.append(": ");  
        contents.append(Thread.currentThread().getName());  
        contents.append(message);  
        contents.append("\n");  
    }  
  
    public String getContents(){  
        return contents.toString();  
    }  
}
```

- A. THIS CLASS IS ALREADY THREAD SAFE
 - B. REPLACING STRINGBUILDER WITH STRINGBUFFER WILL MAKE THIS CLASS THREAD-SAFE
 - C. SYNCHRONIZED THE LOG METHOD ONLY
 - D. SYNCHRONIZED THE GETCONTENTS METHOD ONLY
 - E. SYNCHRONIZED THE LOG AND GETCONTENTS METHOD ONLY
 - F. THIS CLASS CANNOT BE MADE THREAD-SAFE
- ANS. B

VALID METHODSORTERS PROVIDED BY JUNIT

- A. METHODSORTER.DEFAULT
- B. METHODSORTER.JVM
- C. METHODSORTER.NAME_ASCENDING
- D. METHODSORTER.NAME_DESCENDING

ANS: B, C

THE METHOD IN TESTSUITE CLASS THAT RETURNS THE FAILED TESTS AND LOGS

WARNING MESSAGE IS

- A. STATIC TEST SHOWWARNING(STRING MESSAGE)
- B. STATIC TEST WARNING(STRING MESSAGE)
- C. STATIC TEST DISPLAYWARNING(STRING MESSAGE)
- D. STATIC TEST GETWARNING(STRING MESSAGE)

ANS A.

WHICH METHOD IN TESTRESULT CLASS GIVES THE COUNT OF DETECTED ERRORS?

- A. INT ERRORCOUNT()
- B. INT GETERRORCOUNT()
- C. INT COUNTERERRORS()
- D. INT GETERRORS()

ANS GETFAILURECOUNT/ GETERRORCOUNT

CAN YOU AUTOWIRE BYTYPE WHEN MORE THAN ONE BEAN WITH THE SAME TYPE EXIST?

- A. FALSE
- B. TRUE

ANS. A. FALSE

IMAGINE THAT THERE ARE TWO EXCEPTION CLASSED MYBASEBUSINESSEXCEPTION AND MYSUBBASEBUSINESSEXCEPTION WHICH EXTENDS FROM MYBASEBUSINESSEXCEPTION. IF ROLLBACK-FOR ="MYBASEBUSINESSEXCEPTION" AND NO-ROLLBACK-FOR="MYSUBBASEBUSINESSEXCEPTION" ARE CONFIGURED WHAT HAPPENS WHEN A MYSUBBASEBUSINESSEXCEPTION IS THROW FROM TRANSACTION CONTEXT?

- A. NO-ROLLBACK-FOR RULE IS APPLIED AND TRANSACTION WILL NOT ROLLBACK
- B. ROLLBACK-FOR RULE IS APPLIED AND TRANSACTION WILL BE ROLLBACK
- C. EXCEPTION IS THROWN DURING APPLICATION CONTEXT INITIALIZATION SINCE CONFIGURATION IS AMBIGUOUS
- D. NONE OF THE ABOVE

ANS -A

WHICH MESSAGE CONVERTER IS USED TO CONVERT JSON REPRESENTATION TO OBJECT AND VICE VERSA IN SPRING FRAMEWORK

- A. STRINGHTTPMESSAGECONVERTER
- B. HTTPMESSAGECONVERTER
- C. MAPPINGJACKSON2HTTPXMLMESSAGECONVERTER
- D. MAPPINGJACKSON2HTTPMESSAGECONVERTER

ANS.D

WHICH AMOUNG THE FOLLOWING ARE THE POPULAERLY USED APPROCHES TO VERSIONING A SPRING REST API

- A. URI PARAMETER
- B. CONTROLLER HEADER

- C. ACCEPT HEADER
- D. CUSTOM HEADER
- E. ALL
- F. NONE

ANS ACD

THE ANNOTATION IN JAX-RS API ASSIGNS A DEFAULT VALUE TO A PARAMETER PASSED TO METHOD

- A. @FORMPARAM B. @COOKIEPARAM C. @DEFAULTVALUE D. @CONTEXT
- ANS. C

FUNCTIONAL INTERFACE

WHAT IS THE BENEFITS OF USING GENERICS IN YOUR CODE?

- A. IMPROVE PERFORMANCE
- B. MAKE CODE MORE READABLE
- C. ADD STABILITY TO CODE BY MAKING ERROR DETECTABLE AT COMPILE TIME
- D. ADD STABILITY TO CODE BY MAKING ERROR DETECTABLE AT RUNTIME TIME

ANS B,C

WHICH OF THE FOLLOWING IS A BENEFIT FROM USING @OVERRIDE ANNOTATION?

- A. OVERRIDE ANNOTATION SAYS THAT THE METHOD MARKED MUST BE OVERRIDEN BY SUBCLASS
 - B. IT HELPS IN RUNTIME TO DYNAMIC BINDING THE METHOD TO APPROPRIATE OBJECT
 - C. COMPILER CAN WARN YOU IF THE METHOD DESIGNED WITH OVERRIDE IS NOT A CORRECT OVERRIDE
 - D. NONE OF THE ABOVE
- ANS -C

BELOW REFERENCES WILL BE FINALIZED BUT MEMORY WILL NOT BE RECLAIMED AND USEFUL IN THE SCENARIO WHERE NOTIFICATION IS REQUIRED BEFORE OBJECT IS COLLECTED?

- A. STRONG REFERENCE B. SOFT REFERENCE C. WEAK REFERENCE D. PHANTOM REFERENCE
- ANS. D

METHOD OF RUNTIME CALL FORCIBLY TERMINATES THE CURRENTLY RUNNING JVM == RUNTIME.GETRUNTIME.EXIT

THREAD POOL CAN BE CREATED USING A. CALLABLE INTERFACE B. FUTURE INTERFACE C. EXECUTORS D. NONE OF THESE

ANS C

WHICH ONE IS TRUE

- A. A THREAD CAN CALL A NON-SYNCHRONIZED METHOD OF AN OBJECT WHEN A SYNCHRONIZED METHOD IS BEING EXECUTED
- B. THREAD.SLEEP RELEASES THE LOCK
- C. TWO THREADS CAN CALL TWO DIFFERENT SYNCHRONIZED INSTANCE METHODS OF AN OBJECT
- D. WAIT KEEPS THE LOCKS

ANS. A

WHICH OF THE FOLLOWING IS ALSO CALLED HAS-A RELATIONSHIP. A. COMPOSITION B. ASSOCIATION C. AGGREGATION D. NONE OF THE ABOVE

ANS. C AGGREGATION

WHEN EXCEPTION CAN OCCUR IN JAVA CODE RUN TIME/COMPILE TIME - RUNTIME
WHICH OF THE FOLLOWING IS NOT JAVA LEGAL IDENTIFIER GOTO, UNSIGNED,
\$STRING, _XYZ - GOTO

THE ADVANTAGE OF _____ IS THAT THEY SOLVE THE PROBLEM OF SEQUENTIAL
STORAGE REPRESENTATION. BUT DISADVANTAGE IN THAT IS THEY ARE SEQUENTIAL
LISTS.

A. LIST B. LINKED LIST C. TREE D. QUEUE - ANSWER B

HOW DOES SET ENSURE THERE ARE NO DUPLICATES? HASHCODE THEN EQUALS

Chain of Responsibility : WHICH PATTERN TO USE WHEN MORE THAN ONE OBJECT CAN
HANDLE A REQUEST AND THE HANDLER IS UNKNOWN

Command Design Pattern: WE HAVE TO DESIGN AN EDITOR AND GIVE USER DIFFERENT
WAYS TO OPEN A FILE LIKE OPEN BY MENU, OPEN BY TOOL AND OPEN BY EXPLORER,
WHICH DESIGN PATTERN IS IT

Factory Method: CALENDAR CLASS IS AN ABSTRACT CLASS THE GETINSTANCE
METHOD INSTANTIATES GREGORIAN CALENDAR WITH DEFAULT ZONE AND LOCALE,
WHICH DESIGN PATTERN IS IT

THREAD A IS NOTIFIED OR AFTER 2 SEC

ASSUME THE FOLLOWING METHOD IS PROPERLY SYNCHRONIZED AND CALLED FROM
A THREAD A ON AN OBJECT B: WAIT(2000); AFTER CALLING THIS METHOD WHEN WILL
THE THREAD A BECOME A CANDIDATE TO GET ANOTHER TURN AT THE CPU?

IN-PLACE: ALGO LIKE QUICK SORT DOESN'T REQUIRE EXTRA MEMORY FOR
CARRYING OUT THE SORTING PROCEDURE, THIS TECHNIQUE IS CALLED

1. D
2. A
3. D
4. D
5. C
6. A
7. C
8. B
9. A
10. C
11. A
12. B
13. C
14. A
15. A
16. A
17. A
18. D
19. B
20. D
21. B

- 22. D
- 23. C
- 24. A
- 25. B
- 26. B
- 27. A
- 28. D
- 29. B
- 30. A
- 31. C
- 32. D
- 33. C
- 34. A
- 35. B
- 36. A
- 37. A
- 38. D
- 39. D
- 40. D

Linked lists are data structures that are best suited for scenarios where data size is unpredictable or frequently changing. They are flexible and dynamic, and can efficiently handle insertions and deletions at arbitrary positions.

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To create a new map with the same iteration order as an existing map, you should use the LinkedHashMap concrete implementation of the Map interface.

Here's why:

HashMap:

Prioritizes speed and doesn't maintain any specific order for its elements.

TreeMap:

Orders keys by their natural ordering or a specified Comparator, resulting in a sorted iteration order different from the original map unless it was already sorted.

LinkedHashMap:

Maintains the insertion order of elements, ensuring the new map iterates through elements in the same order as the existing map. This makes it ideal for preserving the original iteration order.

Therefore, if you need to preserve the existing iteration order, LinkedHashMap is the best option. It offers performance close to HashMap while maintaining the desired order.

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Which methods may cause a thread to stop executing?

wait()
notify()

notifyall()
exits synchronized code
Answer A. wait().

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(more)

1. The wait() may cause a thread to stop executing.

(more)

The wait() may cause a thread to stop executing.

Solution

Option A is correct. The wait() method makes the active thread wait until another thread calls this object's notify() or notifyAll() methods.

Option B is incorrect. A single thread waiting on this object's monitor is awakened by the notify() function.

Option C is incorrect. Notifies every thread that is waiting on this object's monitor by calling notifyAll().

Option D is incorrect. The thread holding the lock or the thread in the synchronized method typically quits the synchronized method when a lock is released. The lock is released when another thread enters a synchronized procedure on that object. Does executing threads halt while entering or leaving synchronized code? Not required, as the thread can still execute unsynchronized code. The word directly in the question, in my opinion, gives us a hint. The execution of a thread is not immediately stopped when synchronized code is exited.